

4 (long)

(a) Show that the coordinate transformation $\mathbf{U}$ mapping (x, y) into (x', y') such that

$$x' = x \cos \theta + y \sin \theta$$

$$y' = -x \sin \theta + y \cos \theta$$

with $\theta = 45^\circ$, maps the corners of a square defined by $(1, 1)$, $(1, 2)$, $(2, 2)$, $(2, 1)$ into the corners of another square with the same area. [10]

(b) (i) Show that $\mathbf{U}^4 = -\mathbf{I}$, where $\mathbf{I}$ is the identity matrix

(ii) Show that $\mathbf{U}^{-1} = \mathbf{U}^T$. [5]

(c) If

$$\mathbf{A} = \begin{bmatrix} 0 & a & a \\ a & 0 & a \\ a & a & 0 \end{bmatrix}$$

and $a > 0$, find the three eigenvalues of $\mathbf{A}$ and the normalised eigenvector associated with the highest eigenvalue. What can one say about the direction of the other two eigenvectors? [15]